Your grade: 100%

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Next item

Instructions

1. A farmer has two types of fruit baskets: apple baskets and orange baskets. He wants to determine the weight of each type of basket. The farmer performs two weightings: 1 / 1 point

1. He weighs three apple baskets and one orange basket and gets a total weight of 1400 grams.
2. He weighs one apple basket and three orange baskets and gets a total weight of 1200 grams.

Which of the following linear systems describes the experiment above?

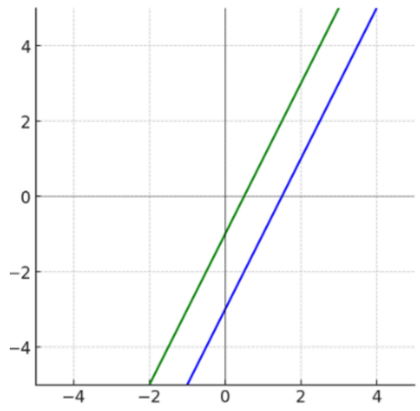
- [x] $\begin{cases} 3a + o = 1400 \\ a + 3o = 1200 \end{cases}$

Nice work

Well done! You correctly identified the system of equations that describes the problem. The equations 3a+o=1400 and a+3o=1200 perfectly represent the two situations: three apple baskets and one orange basket weighing 1400 grams, and one apple basket and three orange baskets weighing 1200 grams.

- [ ] $\begin{cases} 3a = 1400 \\ 3o = 1200 \end{cases}$
- [ ] $\begin{cases} 3a + o = 1200 \\ a + 3o = 1400 \end{cases}$
- [ ] $\begin{cases} 3a + 3o = 1400 \\ 3a + 3o = 1200 \end{cases}$

2. The graph shows two parallel lines. Which system of equations matches this plot? 1 / 1 point

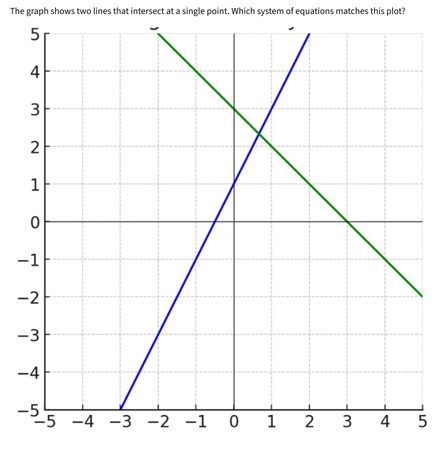


- [ ] $\begin{cases} x + y = 3 \\ y = 1 \end{cases}$
- [ ] $\begin{cases} x - 2y = 2 \\ x + 2y = 5 \end{cases}$
- [x] $\begin{cases} 2x - y = 3 \\ 2x - y = 1 \end{cases}$
- [ ] $\begin{cases} x + y = 4 \\ 3x - y = 2 \end{cases}$

Nice work

Excellent! You correctly identified the system of equations. The two lines are parallel because they have the same slope but different intercepts, which means they will never intersect.

3. The graph shows two lines that intersect at a single point. Which system of equations matches this plot? 1 / 1 point



- [ ] $\begin{cases} x - y = 2 \\ y = \frac{1}{2}x + 1 \end{cases}$
- [ ] $\begin{cases} y = -x + 1 \\ y = 2x + 3 \end{cases}$
- [ ] $\begin{cases} x + 2y = 5 \\ y - x = 1 \end{cases}$
- [x] $\begin{cases} 2x - y = -1 \\ x + y = 3 \end{cases}$

Nice work

Great job! The system of equations you selected is correct. The two lines intersect because they have different slopes, resulting in a single intersection point.

4. Given the system of linear equations: 1 / 1 point

$$\begin{aligned} 3x + 4y &= 10 \\ 6x - 2y &= 5 \end{aligned}$$

Which of the following represents the correct matrix form for this system?

- [ ] $\begin{bmatrix} 6 & 4 \\ -3 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 10 \\ 5 \end{bmatrix}$
- [ ] $\begin{bmatrix} 3 & -4 \\ 2 & 6 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 10 \\ 5 \end{bmatrix}$
- [x] $\begin{bmatrix} 3 & 4 \\ 6 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 10 \\ 5 \end{bmatrix}$
- [ ] $\begin{bmatrix} 3 & -4 \\ 6 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 10 \\ 5 \end{bmatrix}$

Nice work

Excellent! You correctly identified the matrix form. The coefficients of x and y from each equation form the matrix, and the constants on the right side make up the right-hand vector.